MATH MA'AM ACADEMY

Where Every Subject Clicks.

DR. HOMI BHABHA BALVAIDNYANIK COMPETITION | STD. 6

UNIT 6: FOOD & NUTRITION

Compact Study Notes

1. Nutrients and their functions

• **Nutrients** are chemical substances in food that provide energy and materials needed for growth, repair and normal functioning of the body. The main groups are carbohydrates, proteins, fats, vitamins, minerals, roughage and water.

• **Carbohydrates:** Main energy-giving nutrients. Sources include cereals, rice, wheat, potatoes and bananas.

• **Proteins:** Body-building nutrients required for growth and repair. Sources include pulses, beans, milk, meat, fish and eggs.

• **Fats:** Concentrated source of energy; also help insulation and absorption of fat-soluble vitamins. Sources include oils, ghee, butter, nuts and cheese.

• **Vitamins and minerals:** Protective and regulatory nutrients. Sources include fruits, green leafy vegetables, milk, eggs, meat and fish.

• **Roughage (dietary fibre):** Mainly comes from plant foods such as whole grains, fruits and vegetables. It adds bulk to food and helps bowel movement.

• **Water:** Acts as a solvent, helps transport nutrients and wastes, supports chemical reactions and helps regulate body temperature.

2. Balanced diet

• A **balanced diet** contains all essential nutrients in suitable amounts according to a person's age, sex, activity level and health condition.

• A balanced diet should include a variety of foods rather than relying on one food group.

• Requirements vary with growth, physical activity, pregnancy, illness and other physiological conditions.

3. Deficiency diseases

• Deficiency diseases occur when the diet lacks one or more essential nutrients over a period of time.

• **Kwashiorkor:** Severe protein deficiency, especially in children; may cause a swollen belly, stunted growth, muscle wasting, skin changes and diarrhoea.

• **Marasmus:** Severe deficiency of energy and protein; the child becomes extremely thin and loses body fat and muscle.

• **Goitre:** Often caused by iodine deficiency; the thyroid gland enlarges in the neck. Iodine is needed to make thyroid hormones.

• **Beri-beri:** Due to vitamin B1 (thiamine) deficiency; can cause weak muscles and nerve-related problems.

• **Scurvy:** Due to vitamin C deficiency; causes swollen or bleeding gums and slow wound healing. Citrus fruits are useful sources of vitamin C.

• **Anaemia:** Iron deficiency is a common cause; reduced haemoglobin can lead to paleness, tiredness and weakness. Anaemia can have other causes too.

• **Rickets:** In children, commonly associated with vitamin D deficiency; bones become soft and may bend or deform. Vitamin D supports calcium absorption and bone mineralisation.

• **Night blindness:** Can result from vitamin A deficiency; difficulty seeing in dim light.

• **Pellagra:** Due to niacin (vitamin B3) deficiency; classically associated with dermatitis, diarrhoea and dementia.

• **Kwashiorkor and marasmus** are both forms of severe malnutrition, but their typical presentations differ.

4. Food preparation and cooking

• Some foods are eaten raw, while others need cooking. Cooking improves flavour, texture and digestibility and can destroy many harmful microbes.

• Overheating, overcooking and repeated washing of cut vegetables or pulses can cause nutrient loss.

• Use suitable cooking methods, avoid excessive water when possible, and do not discard nutrient-rich cooking water unnecessarily.

5. Food preservation methods

• **Heating:** High temperature kills many microbes; heating is used in food processing, including milk treatment.

• **Dehydration:** Removing water by drying slows microbial growth. Examples include dried fruits, grains and papad.

• **Pasteurisation:** Controlled heating followed by rapid cooling reduces harmful microorganisms in liquids such as milk. Exact time-temperature combinations depend on the process; the textbook gives heating to around 70°C followed by rapid cooling.

• **Smoking:** Used especially for meat and fish; combines drying with compounds from smoke that help preservation.

• **Refrigeration:** Low temperature slows microbial multiplication and food spoilage; it does not sterilise food.

• **Canning:** Food is processed and sealed in airtight containers to prevent re-entry of microorganisms.

• **Preservatives:** Salt, sugar, oil, vinegar and approved chemical preservatives such as sodium benzoate may be used in suitable foods.

• **Neem leaves:** Traditionally placed with stored grains to help protect them from insect pests. Use dry, clean storage and approved food-safe methods.

6. Important food tests and facts

• **Starch test:** Iodine solution turns blue-black in the presence of starch. Potato is rich in starch; sugar does not give the blue-black starch reaction.

• Water does not provide energy like carbohydrates and fats. It is essential as a solvent, transport medium and participant in body processes.

• Food preservation aims to delay spoilage by controlling microorganisms, moisture, temperature, air or pests.

7. Quick revision table: deficiency and nutrient

Deficiency	Likely condition
Protein	Kwashiorkor; severe overall energy/protein deficiency can cause marasmus
Iodine	Goitre
Vitamin B1	Beri-beri
Vitamin C	Scurvy
Iron	Anaemia (common cause)
Vitamin D	Rickets in children
Vitamin A	Night blindness
Niacin (B3)	Pellagra

ALL THE BEST